

Ashvin K S

COMPUTER SCIENCE UNDERGRADUATE | AI SYSTEMS & WEB DEVELOPER

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EDUCATION

Vellore Institute of Technology, Chennai

Aug 2024 – May 2028

B.Tech — Computer Science & Engineering

Current CGPA: 9.26 / 10.00

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Design & Analysis of Algorithms, Computer Architecture

TECHNICAL SKILLS

Languages	TypeScript, Python, Rust, JavaScript, Java, SQL, C, Shell/Bash
Frontend	React, Next.js, Vite, Tailwind CSS, Framer Motion, HTML/CSS
Desktop	Tauri, Rust IPC
Backend	Node.js, FastAPI, Flask, MySQL, MongoDB
AI / ML	PyTorch, NVIDIA NIM APIs, HuggingFace, CUDA Python
DevOps	GitHub Actions, Vercel, Docker, AWS, Google Cloud

PROJECTS

Atheletia — Desktop AI Workspace

TypeScript · Tauri · NVIDIA NIM ↗ GitHub

- Built a local-first desktop productivity hub in Tauri + React + Rust that unifies AI chat, a notes vault with local retrieval (RAG), activity tracking, calendar, and music into one app.
- Wrote a Rust IPC layer that handles filesystem access and SQLite storage so all data stays on-device with no network calls.

DuckGraph — Code Dependency Graph Visualizer

TypeScript · tree-sitter · SQLite ↗ GitHub

- Built a VS Code extension that uses tree-sitter to parse source files and builds a SQLite-backed dependency graph for cross-file navigation and symbol-level queries.
- Ships an Express daemon plus an MCP server that lets AI coding agents query the graph directly — find symbols, trace connections, read source context — instead of grepping files.

Amorvin Chess AI — Deep RL + MCTS

Python · PyTorch ↗ GitHub

- Trained a CNN with 12 residual blocks (AlphaZero-style) on Lichess PGN data via supervised learning, then fine-tuned with self-play reinforcement learning.
- Implemented MCTS with UCB1 selection, Dirichlet noise for exploration, and a dual-head network (policy: move probability distribution over all legal moves; value: position value score).

CERTIFICATIONS

Full Stack Open

University of Helsinki · 2026

Building RAG Agents with LLMs

NVIDIA Deep Learning Institute · 2025

Google Cloud Learning Paths

Google Cloud · 2026

Machine Learning Specialization

DeepLearning.AI · 2025

Fundamentals of CUDA Python

NVIDIA Deep Learning Institute · 2025

ACTIVITIES & LEADERSHIP

GirlScript Summer of Code (GSSoC 2026) — Open Source Contributor

2026

28 merged PRs across 14 open-source repositories — config standards, CI workflows, Docker environments, and contributor tooling.

Full-Stack Developer — Microsoft Innovations Club

2024 – Present

VIT Chennai · Contributing to club web projects, internal tools, APIs, and practical deployment workflows.

Technical Volunteer — CodeChef VITC Chapter

2024 – Present

Supporting algorithmic practice, competitive programming sessions, and campus technical events.

Outreach — Android Club VITC

Oct 2025 – Apr 2026

Reached out to companies and industry contacts to coordinate speakers, sponsorships, and event partnerships for club workshops and hackathons.